Build a React App in 20 Minutes with AI Assistance

From zero to a multi-page React app — with AI by your side

👋 Introduction

https://www.zainfathoni.com/about

- 📍 Jember → Bandung → 🇸🇬 SG → Jogja
- 🛠️ Backend → Manager → Frontend → Fullstack
- 🤖 AI Enthusiast · Speaker · Empowered Father 🚀

🤔 Opening Question

“ How long does it take you to build an app from scratch? ”

📚 Agenda

1. **Why React?** — The modern developer's primary tool
2. **React Fundamentals** — Core concepts including React Router
3. **AI as a Coding Assistant** — A workflow that changes everything
4. **Live Demo** — Building a real app together
5. **Takeaways** — Concrete steps you can start today

→ **Goal:** Go home with new skills + a new workflow

🌍 Why React?

React is used by:

- Meta · Netflix · Airbnb · Shopify · Discord
- Tokopedia · Gojek · Traveloka · Bukalapak

→ **Not just big companies — small teams and freelancers too**

📊 React in the Developer World

Stack Overflow Survey 2024:

- React: most popular web framework **12 years running**

→ Large ecosystem = **more resources, libraries, and answers on the internet**

⚛️ What Exactly Is React?

> React is a **JavaScript library** for building user interfaces (UI) using **reusable components**

Core concepts:

- 🧩 **Component** — Self-contained pieces of UI
- 🔄 **State** — Data that can change
- ⬇️ **Props** — Data passed to components
- 🔁 **Re-render** — UI automatically updates when data changes

React Component — Simple Example

```
// Product Card Component
function ProductCard({ name, price, image }) {
  return (
    <div className="card">
      <img src={image} alt={name} />
      <h2>{name}</h2>
      <p>Rp {price.toLocaleString()}</p>
      <button>Add to Cart</button>
    </div>
  )
}
```

→ One component → can be used hundreds of times with different data

🔄 State — Components That Can Change

```
import { useState } from 'react'

function Counter() {
  const [count, setCount] = useState(0)
  return (
    <div>
      <p>Count: {count}</p>
      <button onClick={() => setCount(count + 1)}>Add</button>
    </div>
  )
}
```

→ `useState` = how React "remembers" things between renders

React App Structure

```
my-app/
├── src/
│   ├── components/     ← Reusable components
│   │   ├── Button.jsx
│   │   └── ProductCard.jsx
│   ├── pages/          ← App pages
│   │   ├── Home.jsx
│   │   └── Cart.jsx
│   ├── App.jsx         ← Main / root component
│   └── main.jsx        ← Entry point
├── public/             ← Static assets
└── package.json        ← Dependencies & scripts
```

🧭 React Router — Navigation Between Pages

```
// src/App.jsx — with React Router
import { BrowserRouter, Routes, Route, Link } from 'react-router-dom'
import Home from './pages/Home'
import Cart from './pages/Cart'
import ProductDetail from './pages/ProductDetail'

function App() {
  return (
    <BrowserRouter>
      <nav>
        <Link to="/">Home</Link>
        <Link to="/cart">Cart</Link>
      </nav>
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/cart" element={<Cart />} />
        <Route path="/product/:id" element={<ProductDetail />} />
      </Routes>
    </BrowserRouter>
  )
}
```

🧭 Nested Routes — Shared Layouts

```
<Routes>
  <Route element={<Layout />}>        {/* shared nav + footer */}
    <Route path="/" element={<Home />} />
    <Route path="/cart" element={<Cart />} />
    <Route path="/product/:id" element={<ProductDetail />} />
  </Route>
  <Route path="/login" element={<Login />} />  {/* no layout */}
</Routes>
```

```
const Layout = () => (<>
  <Navbar />
  <Outlet />   {/* child route renders here */}
  <Footer />
</>)
```

🧭 Dynamic Routes & useParams

```
// src/pages/ProductDetail.jsx
import { useParams } from 'react-router-dom'
import { products } from '../data/products'

function ProductDetail() {
  const { id } = useParams()
  const product = products.find(p => p.id === Number(id))

  if (!product) return <p>Product not found</p>

  return (
    <div>
      <img src={product.image} alt={product.name} />
      <h1>{product.name}</h1>
      <p>{formatRupiah(product.price)}</p>
    </div>
  )
}
```

📱 Expo
📱 React Native
🔀 Next.js
🔀 React Router
🔀 Remix
🗃️ Redux Toolkit
📡 React Query
⚡ Vite
⚛️ React Core
🗃️ Zustand
📡 SWR
⚡ Webpack
🗃️ Jotai
⚡ Turbopack
🧪 Vitest
🧪 React Testing Library
🎨 Tailwind CSS
🎨 CSS Modules
🎨 styled-components

🤖 Enter: AI as a Coding Assistant

Before AI:

- Open docs → Read → Try → Error → Google → Stack Overflow → Try again

With AI:

- Describe what you want → Get a code draft → Review → Adjust → Done

→ **AI doesn't replace understanding — AI removes friction**

🛠️ AI Tools for React Developers

Tool	Best for	Free?
GitHub Copilot	Autocomplete in editor	Free plan
Claude Code	Refactoring, analysis, multi-file	Paid (API/Subs)
Cursor	AI-native editor	Free tier
v0 by Vercel	Generate UI from description	Free tier
ChatGPT	Q&A, debugging, explanations	Free

→ **Pick one, master it first — don't jump around**

💬 The Art of Talking to AI

Bad prompt:

> "Make a React app"

Good prompt:

> "Create a React component called `ProductList` that takes an array of `products` as props, displays them in a 3-column grid, with loading state and empty state. Use Tailwind CSS."

→ **Specific + context + expected output = accurate results**

🔄 AI-Assisted React Workflow

```
1. DEFINE   → Tell AI what you want to build
      ↓
2. GENERATE → AI creates the first draft
      ↓
3. REVIEW   → Read and understand every line
      ↓
4. REFINE   → Ask AI to fix what's lacking
      ↓
5. TEST     → Run it, see the results
      ↓
6. ITERATE  → Repeat until satisfied
```

→ You're still the pilot — AI is the co-pilot

📋 Plan Mode — Think Before You Code

> "I want to build a shopping cart app with React and React Router. Plan the component structure, routes, and state management before writing any code."

AI will outline:

- Component tree and responsibilities
- Route structure (`/`, `/cart`, `/product/:id`) and shared states
- Step-by-step implementation order

→ **Plan first, code second — fewer rewrites, better architecture**

🎯 What We're Building Today

🛒 Mini Shopping App

- Product list from dummy data
- Shopping cart with state management
- Automatic price total
- Responsive layout

→ **From zero to running in ~20 minutes with React + AI**

Step 1 — Project Setup

Prompt to AI:

> "Create a new React project with Vite and React Router. Provide the setup steps for a shopping cart app with multiple pages."

```
# What AI will guide you through:
npm create vite@latest mini-shop -- --template react
cd mini-shop
npm install react-router-dom
npm run dev
```

Step 2 — Product Data

> "Create a file `src/data/products.js` containing an array of 6 dummy products with fields: id, name, price (in Rupiah), image (from picsum.photos), and category."

```
export const products = [
  {
    id: 1,
    name: "Laptop Gaming Pro",
    price: 15000000,
    image: "https://picsum.photos/seed/laptop/300/200",
    category: "Electronics"
  },
  // ... 5 more products
]
```

Step 3a — ProductCard Prompt

Prompt to AI:

> "Create a `ProductCard` component with image, name, price in Rupiah, an 'Add to Cart' button, and a Link to `/product/:id`. Use inline styles."

Step 3b — ProductCard Result

```
// src/components/ProductCard.jsx — AI output
import { Link } from 'react-router-dom'

function ProductCard({ product, onAddToCart }) {
  return (
    <div style={{ border: '1px solid #e2e8f0', borderRadius: 8 }}>
      <Link to={`/product/${product.id}`}>
        <img src={product.image} alt={product.name} />
      </Link>
      <h3><Link to={`/product/${product.id}`}>{product.name}</Link></h3>
      <p>{formatRupiah(product.price)}</p>
      <button onClick={() => onAddToCart(product)}>Add to Cart</button>
    </div>
  )
}
```

Step 4a — Routes & Cart Prompt

Prompt to AI:

> "Set up App.jsx with BrowserRouter, cart state, and routes for Home, Cart, and Product Detail pages."

Step 4b — Routes & Cart Result

```
// src/App.jsx — AI output
import { useState } from 'react'
import { BrowserRouter, Routes, Route, Link } from 'react-router-dom'

export default function App() {
  const [cart, setCart] = useState([])
  const addToCart = (product) => { /* handle duplicates */ }

  return (
    <BrowserRouter>
      <nav>
        <Link to="/">🏠 Home</Link>
        <Link to="/cart">🛒 Cart ({cart.length})</Link>
      </nav>
      <Routes>
        <Route path="/" element={<Home onAddToCart={addToCart} />} />
        <Route path="/cart" element={<Cart cart={cart} />} />
        <Route path="/product/:id" element={<ProductDetail />} />
      </Routes>
    </BrowserRouter>
  )
}
```

Step 5 — Finishing Touches

Ask AI to:

1. **Product detail page** with `useParams` to show full info
2. **Cart page** with item list, quantities, and total price
3. **Empty state** when cart is empty
4. **Responsive grid** for the product list

→ **AI can handle the polish — you focus on product decisions**

🎉 Demo Results

In ~20 minutes we built:

- ✅ Project setup with Vite + React Router
- ✅ Data layer (dummy products)
- ✅ Reusable component (ProductCard with navigation)
- ✅ Multi-page routing (Home, Cart, Product Detail)
- ✅ Shared state management (shopping cart across routes)

→ **Without AI: estimated 2-3 hours. With AI: 20 minutes.**

⚠️ Pitfalls to Avoid

1. **Not reviewing AI code** → Hidden bugs in production
2. **Over-reliance** → You can't code without AI
3. **Prompts too vague** → Useless output
4. **Skipping tests** → "AI said it's correct" is not a guarantee

🧠 AI as a Tutor

> "Explain useEffect to me like I'm a junior developer just learning React. Give me a real-world example."

→ AI is infinitely patient — it can explain the same thing in different ways

🧠 AI as a Review & Debug Partner

Review:

> "Review this code: what can be improved for performance and React best practices?"

Debug:

> "This code fetches data from an API but causes an infinite loop. Help me debug."

🗺️ React Learning Roadmap

Week 1-2	Week 3-4	Month 2	Month 3+
JSX & Components	State & Props	React Router & API	React Native

AI helps at every stage

→ Consistent 1 hour/day beats an 8-hour marathon once a week

💡 Key Takeaways

1. 🧩 **React** is a worthy foundation — web + mobile, mature ecosystem
2. ⚛️ **Components + State + Props** are the core you must master
3. 🤖 **AI** removes friction, but is not a substitute for understanding
4. 💬 **Good prompts** = accurate results = time saved
5. 🔁 **Review, review, review** — never skip this step

→ **Start today. One component. One prompt. One step.**

🚀 First Steps Tonight

Quick wins (each < 30 minutes):

- ✅ Set up a React project with Vite: `npm create vite@latest`
- ✅ Build your first component: a simple `Button` or `Card`
- ✅ Try one prompt to ChatGPT/Copilot for React
- ✅ Bookmark: react.dev and vitejs.dev
- ✅ Join a community: ReactID, Kelas Terbuka, or Dicoding

→ **Pick 2-3 to do tonight**

🙏 Thank You!

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🔗 **Slides:** zainf.dev/react-with-ai-assistance

💡 *"The best time to start learning React was 5 years ago. The second best time is now — with AI by your side."*